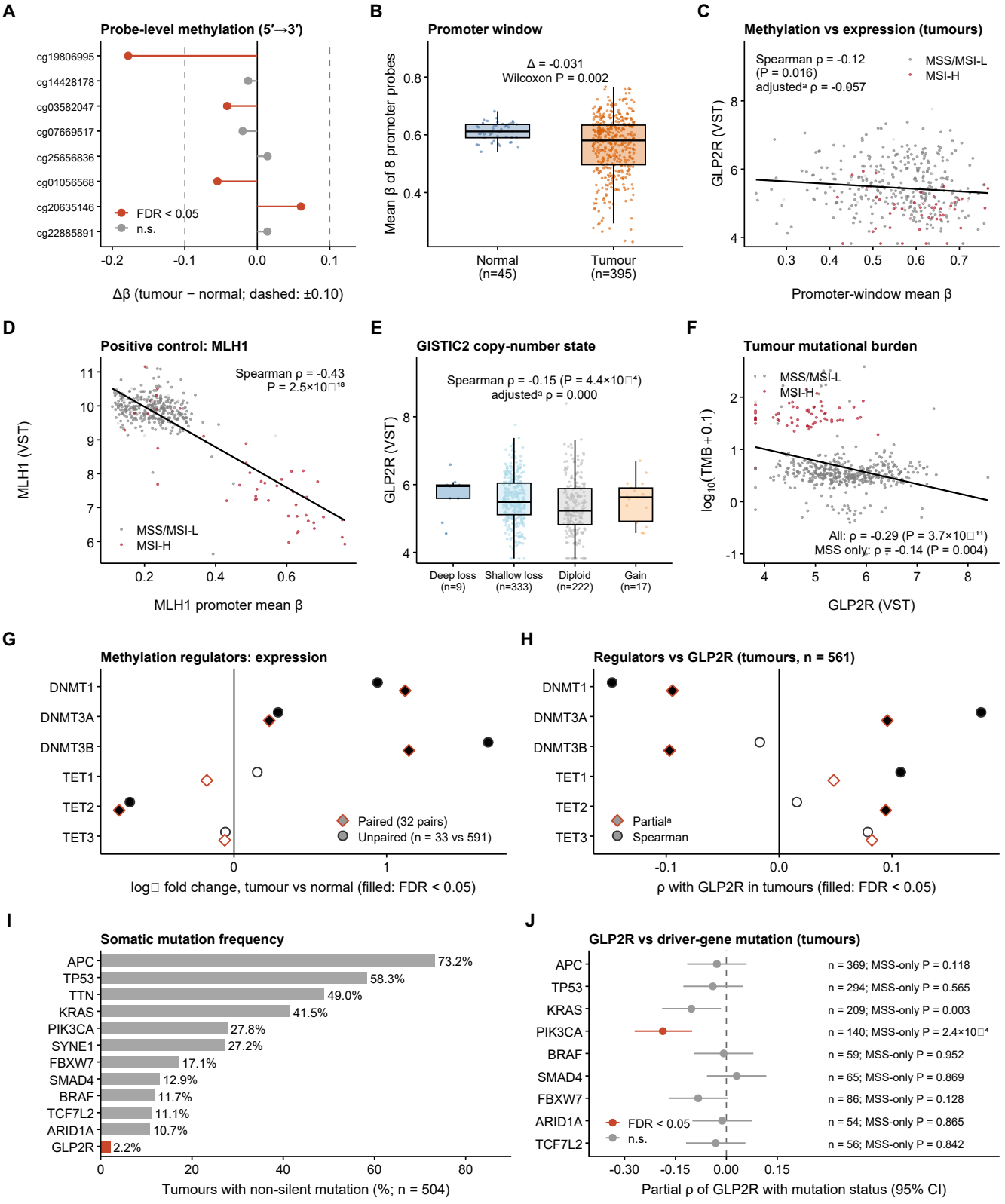

A
Probe-level methylation (5′→3′)
cg19806995
cg14428178
cg03582047
cg07669517
cg25656836
cg01056568
cg20635146
cg22885891
FDR < 0.05
n.s.
-0.2
-0.1
0.0
0.1
Δβ (tumour − normal; dashed: ±0.10)
B
Promoter window
Δ = -0.031
Wilcoxon P = 0.002
Mean β of 8 promoter probes
0.8
0.6
0.4
Normal
(n=45)
Tumour
(n=395)
C
Methylation vs expression (tumours)
Spearman ρ = -0.12
(P = 0.016)
adjusted[a] ρ = -0.057
MSS/MSI-L
MSI-H
GLP2R (VST)
8
6
4
0.3
0.4
0.5
0.6
0.7
Promoter-window mean β
D
Positive control: MLH1
Spearman ρ = -0.43
P = 2.5×10□[18]
MLH1 (VST)
11
10
9
8
7
6
MSS/MSI-L
MSI-H
0.2
0.4
0.6
MLH1 promoter mean β
E
GISTIC2 copy-number state
Spearman ρ = -0.15 (P = 4.4×10□[4])
adjusted[a] ρ = 0.000
GLP2R (VST)
8
6
4
Deep loss
(n=9)
Shallow loss
(n=333)
Diploid
(n=222)
Gain
(n=17)
F
Tumour mutational burden
MSS/MSI-L
MSI-H
log10(TMB + 0.1)
2
1
0
-1
All: ρ = -0.29 (P = 3.7×10□[11])
MSS only: ρ = -0.14 (P = 0.004)
4
5
6
7
8
GLP2R (VST)
G
Methylation regulators: expression
DNMT1
DNMT3A
DNMT3B
TET1
TET2
TET3
Paired (32 pairs)
Unpaired (n = 33 vs 591)
0
1
log□ fold change, tumour vs normal (filled: FDR < 0.05)
H
Regulators vs GLP2R (tumours, n = 561)
DNMT1
DNMT3A
DNMT3B
TET1
TET2
TET3
Partial[a]
Spearman
-0.1
0.0
0.1
ρ with GLP2R in tumours (filled: FDR < 0.05)
I
Somatic mutation frequency
APC 73.2%
TP53 58.3%
TTN 49.0%
KRAS 41.5%
PIK3CA 27.8%
SYNE1 27.2%
FBXW7 17.1%
SMAD4 12.9%
BRAF 11.7%
TCF7L2 11.1%
ARID1A 10.7%
GLP2R 2.2%
0
20
40
60
80
Tumours with non-silent mutation (%; n = 504)
J
GLP2R vs driver-gene mutation (tumours)
APC
TP53
KRAS
PIK3CA
BRAF
SMAD4
FBXW7
ARID1A
TCF7L2
FDR < 0.05
n.s.
n = 369; MSS-only P = 0.118
n = 294; MSS-only P = 0.565
n = 209; MSS-only P = 0.003
n = 140; MSS-only P = 2.4×10□[4]
n = 59; MSS-only P = 0.952
n = 65; MSS-only P = 0.869
n = 86; MSS-only P = 0.128
n = 54; MSS-only P = 0.865
n = 56; MSS-only P = 0.842
-0.30
-0.15
0.00
0.15
Partial ρ of GLP2R with mutation status (95% CI)